trajectories of light rays that begin at some light source, for example, a distant star, and travel inward through Gargantua's warped space and time to the camera. From those light rays, my equations then compute the images the camera sees, taking account not only of the light's sources and Gargantua's warping of space and time, but also the camera's motion around Gargantua.

Having derived the equations, I implemented them myself, using user-friendly computer software called Mathematica. I compared images produced by my Mathematica computer code with Alain Riazuelo's images, and when they agreed, I cheered. I then wrote up detailed descriptions of my equations and sent them to Oliver in London, along with my Mathematica code.

My code was very slow and had low resolution. Oliver's challenge was to convert my equations into computer code that could generate the ultra-high-quality IMAX images needed for the movie.

Oliver and I did this in steps. We began with a nonspinning black hole and a nonmoving camera. Then we added the black hole's spin. Then we added the camera's motion: first motion in a circular orbit, and then plunging into a black hole. And then we switched to a camera around a wormhole.

At this point, Oliver hit me with a minibombshell: To model some of the more subtle effects, he would need not only equations describing the trajectory of a ray of light, but also equations describing how the cross section of a beam of light changes its size and shape during its journey past the black hole.

I knew more or less how to do this, but the equations were horrendously complicated and I feared making mistakes. So I searched the technical literature and found that in 1977 Serge Pineault and Rob Roeder at the University of Toronto had derived the necessary equations in almost the form I needed. After a three-week struggle with my own